MERN

Git and GitHub

Lecture 2

Gaurav Raj

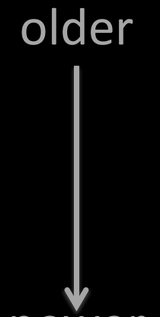
Version Control

Why track/manage different versions of code?

- Backup: Undo or refer to old stuff
- Branch: Maintain old release while working on new
- Collaborate: Work in parallel with teammates

Version Control Systems (VCSs)

- Help you track/manage/distribute revisions
- Standard in modern development
- Examples:
  - Revision Control System (RCS)
  - Concurrent Versions System (CVS)
  - Subversion (SVN)
  - Git



Our focus

GitHub-User Perspective



You



GitHub

GitHub-User Perspective

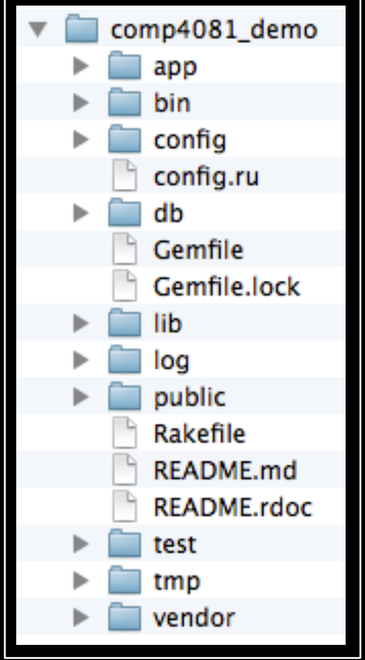


You



GitHub

Working Dir



GitHub-User Perspective

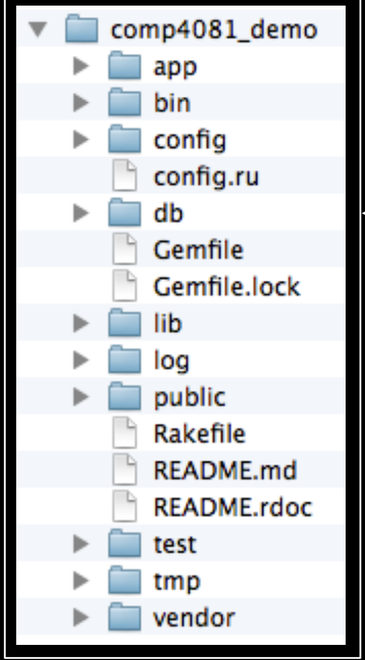


You

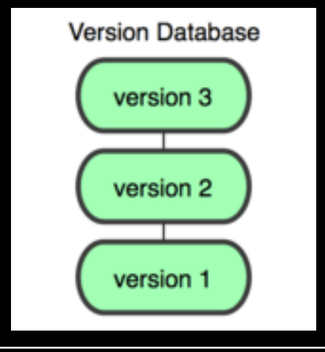


GitHub

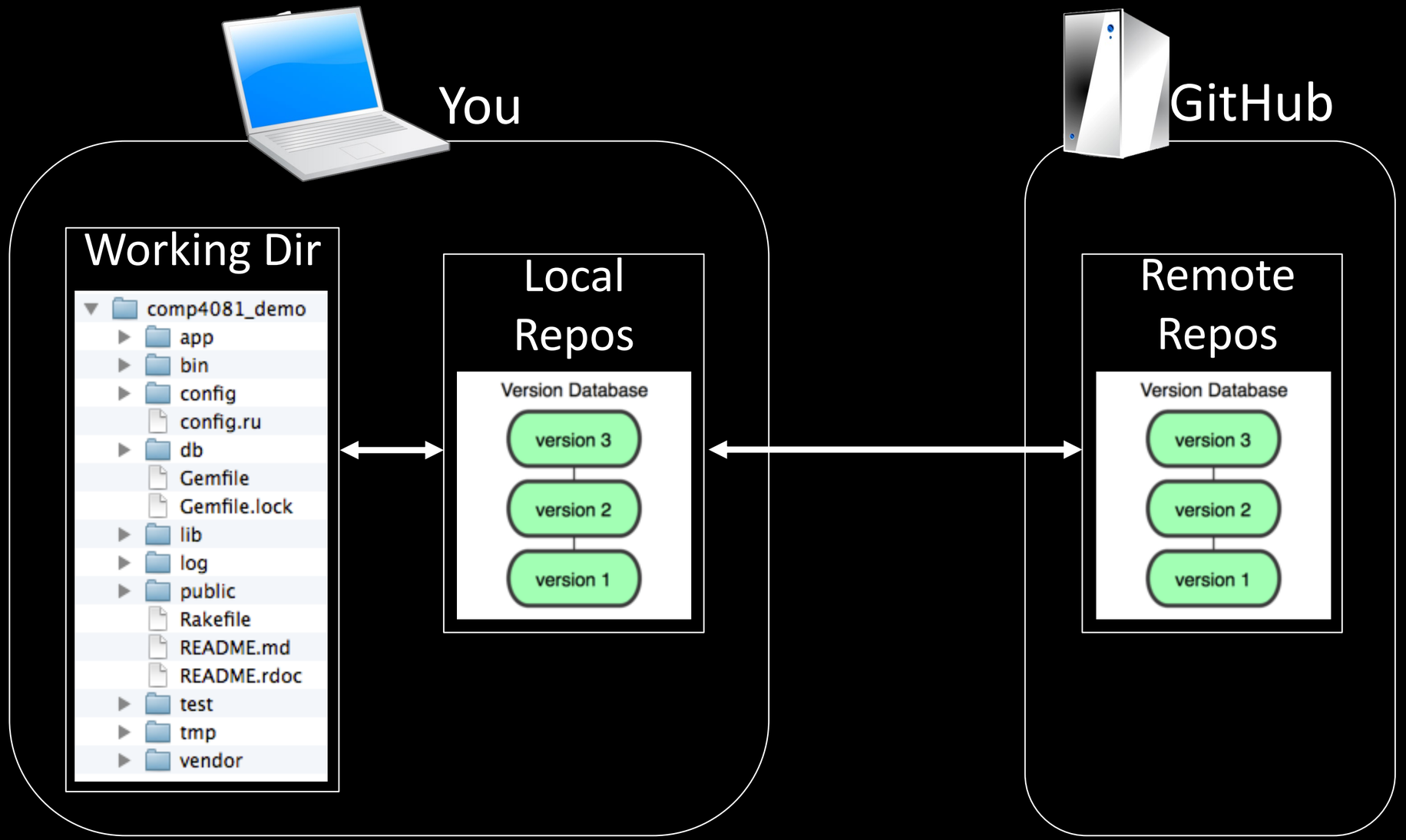
Working Dir



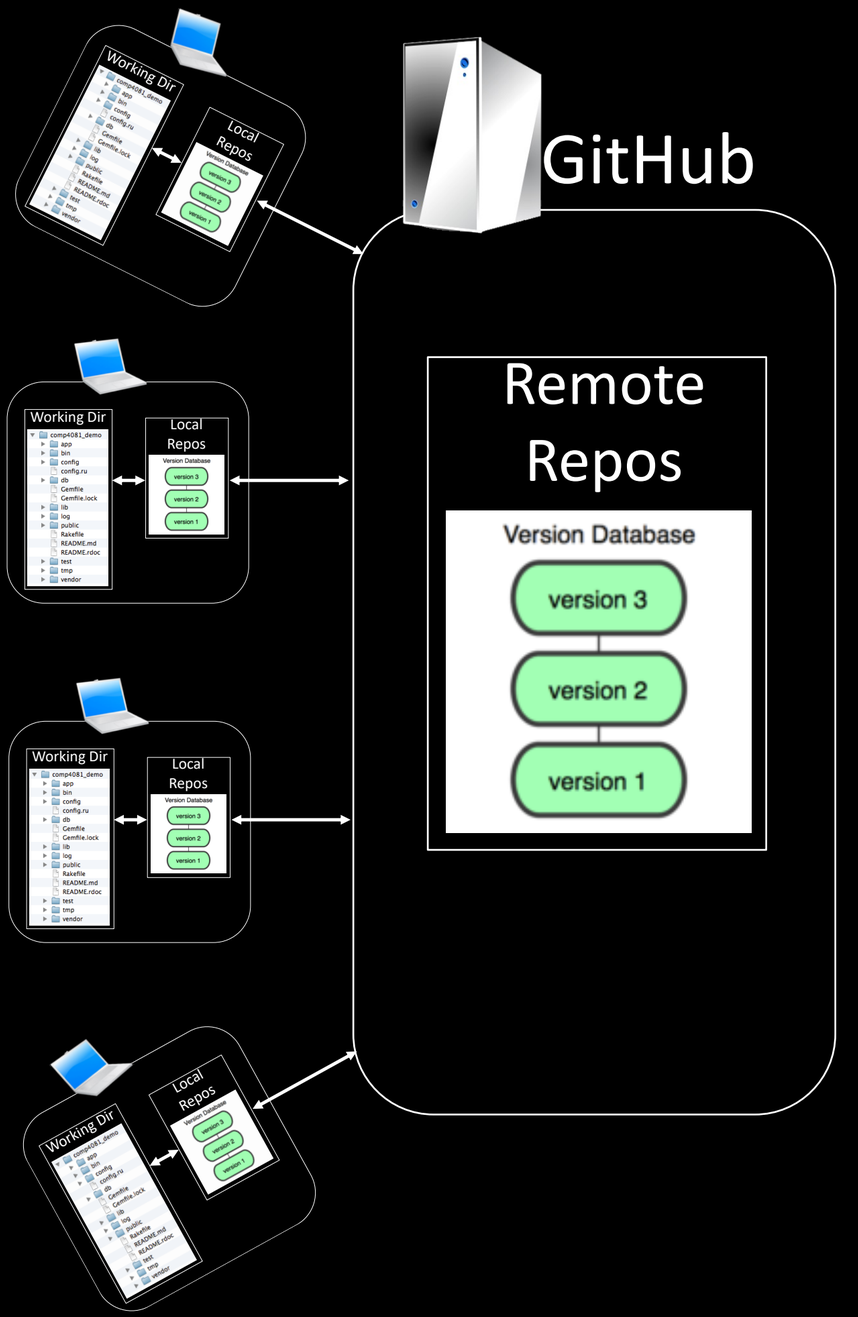
Local Repos



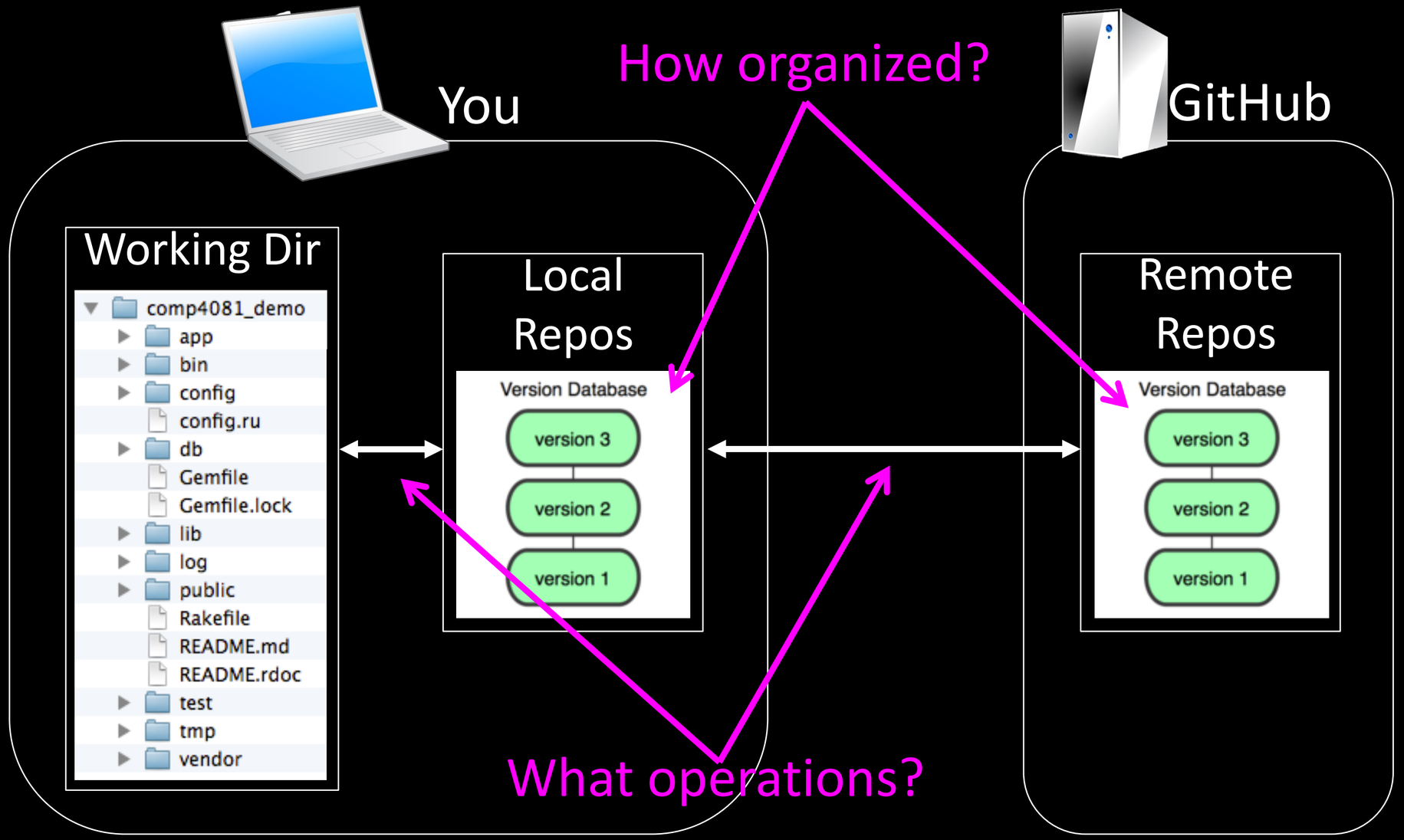
GitHub-User Perspective



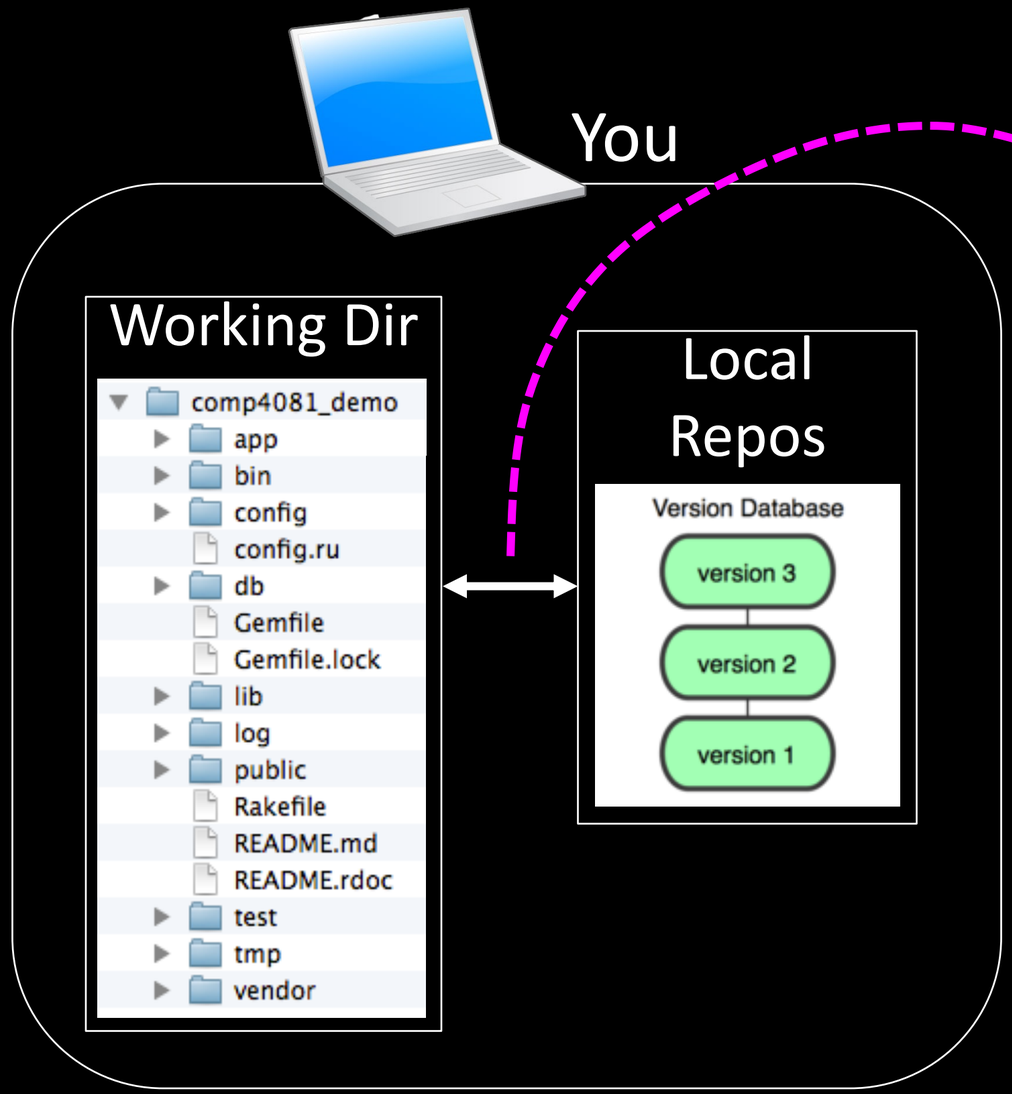
Using GitHub to Collaborate



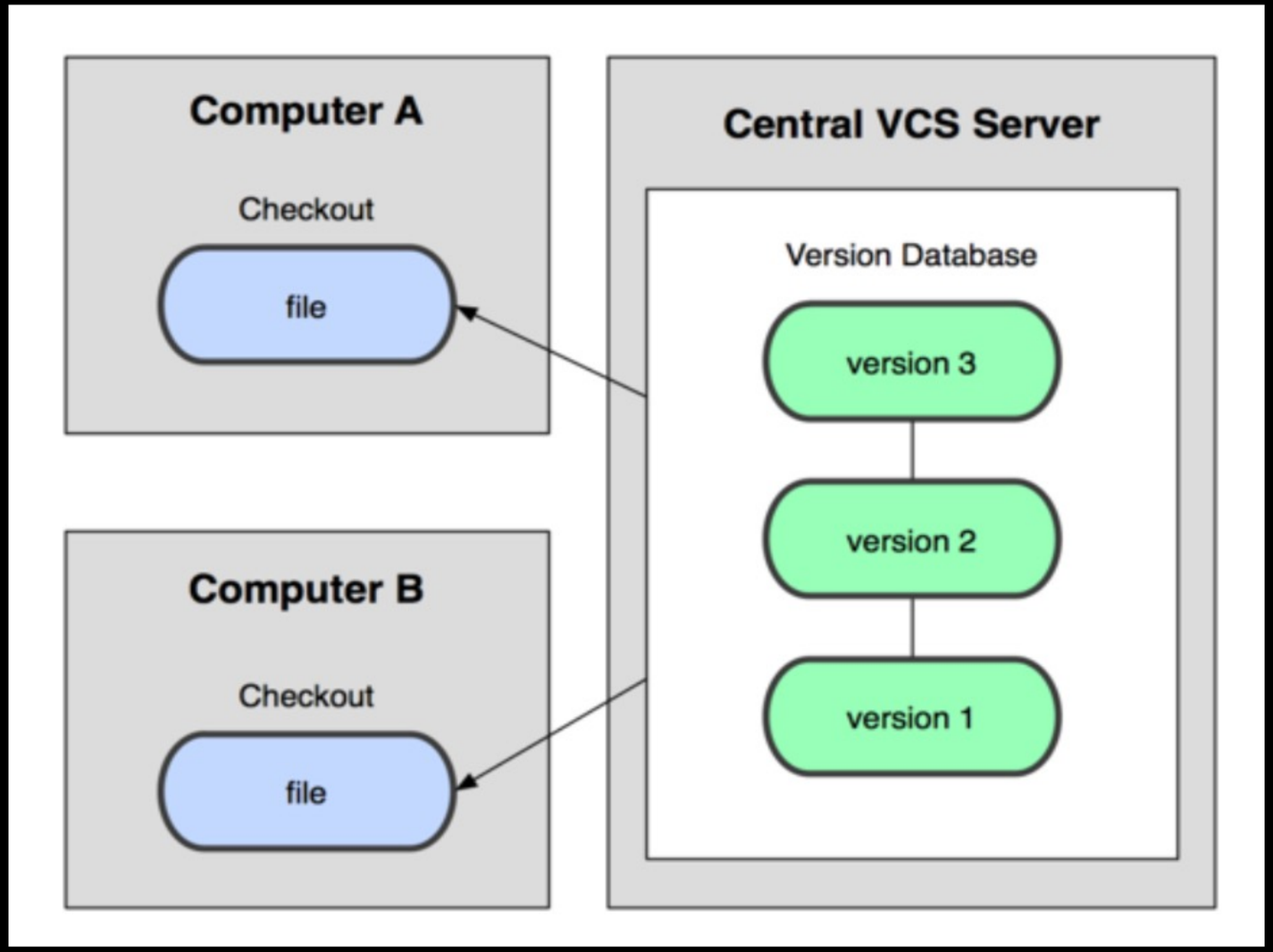
Questions to answer

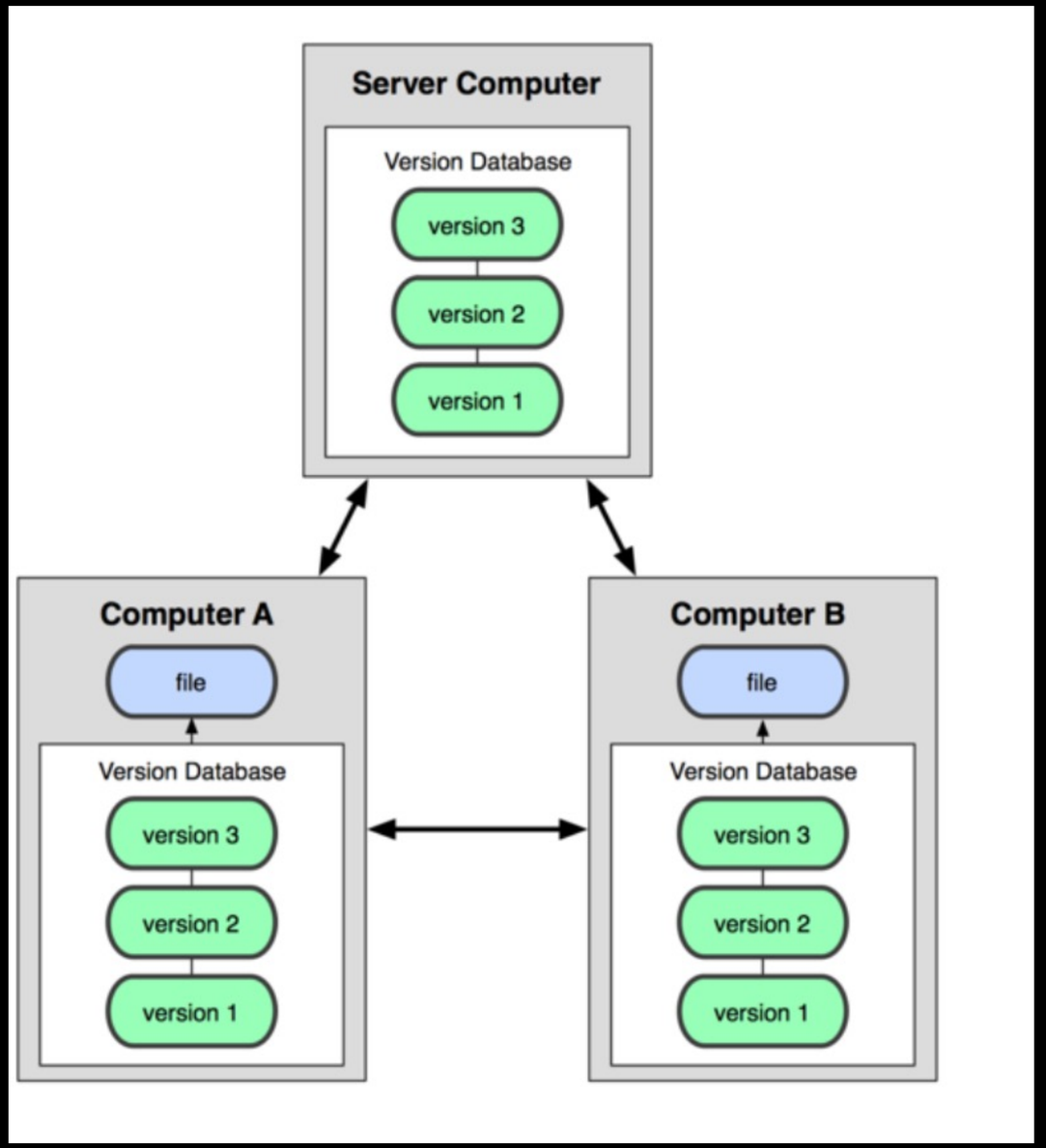


Local Repo Operations

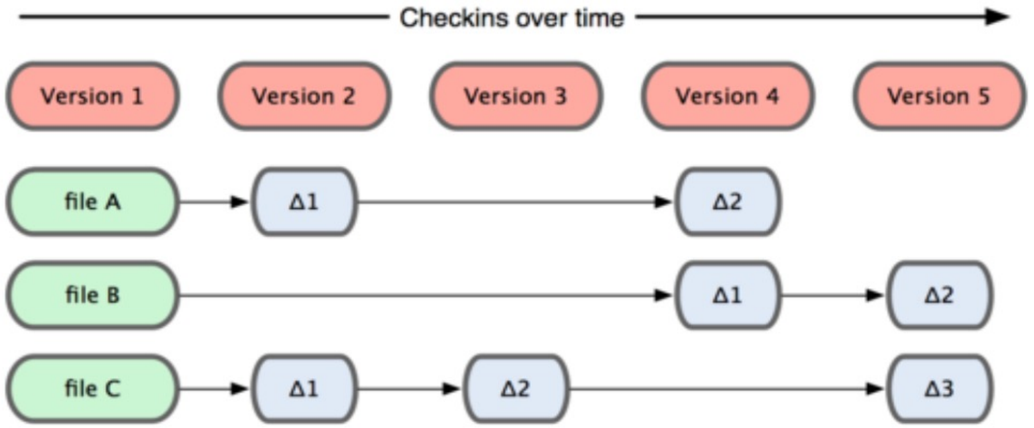


- init
- add/commit
- log
- switch/checkout
- branch
- merge
- …

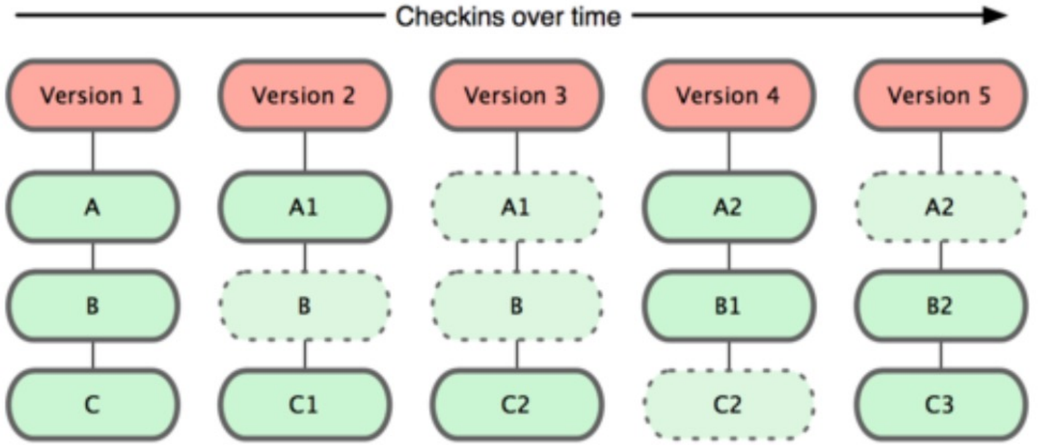

Computer A
Checkout
file
Computer B
Checkout
file
Central VCS Server
Version Database
version 3
version 2
version 1

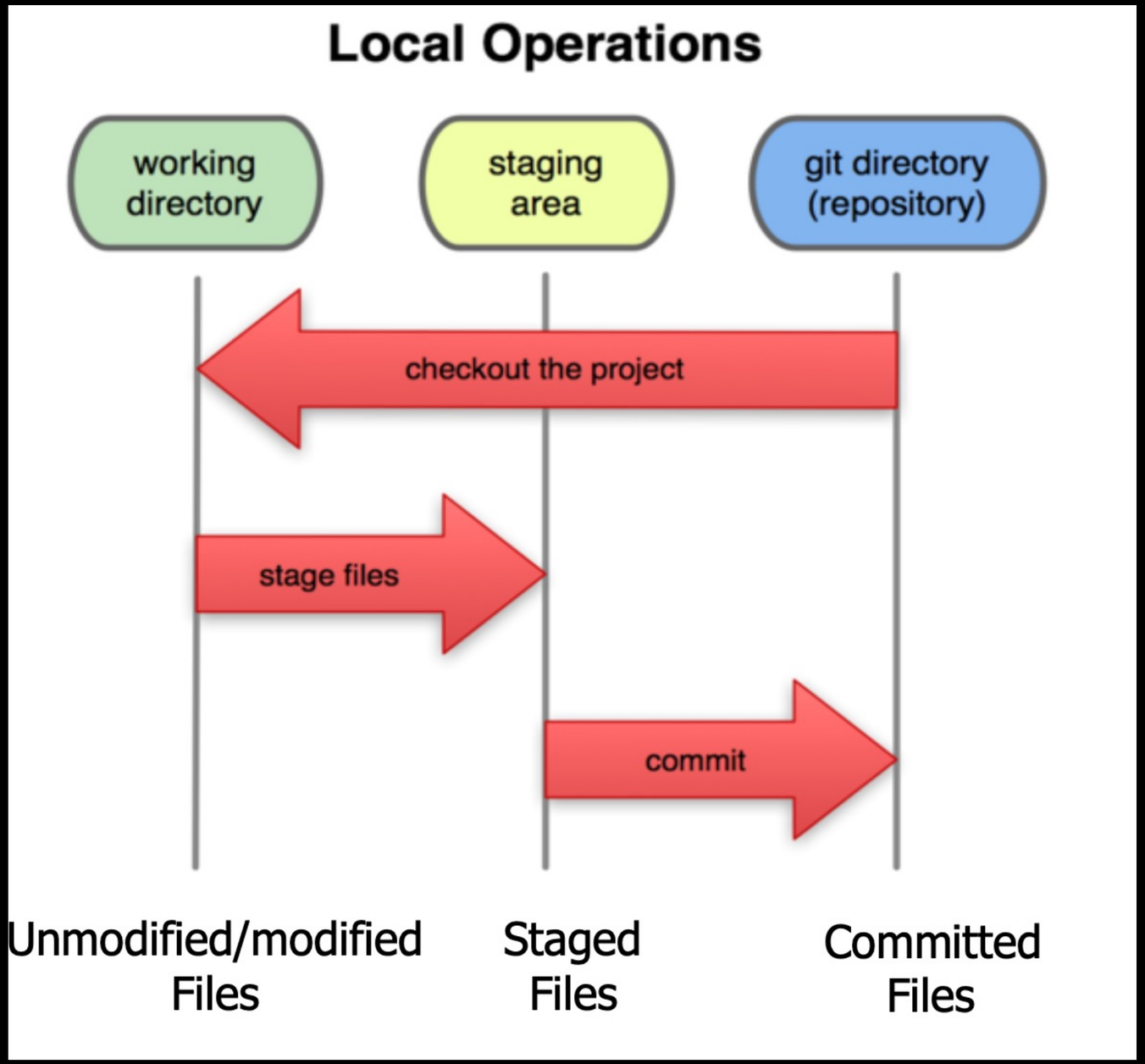

Server Computer
Version Database
version 3
version 2
version 1
Computer A
file
Version Database
version 3
version 2
version 1
Computer B
file
Version Database
version 3
version 2
version 1

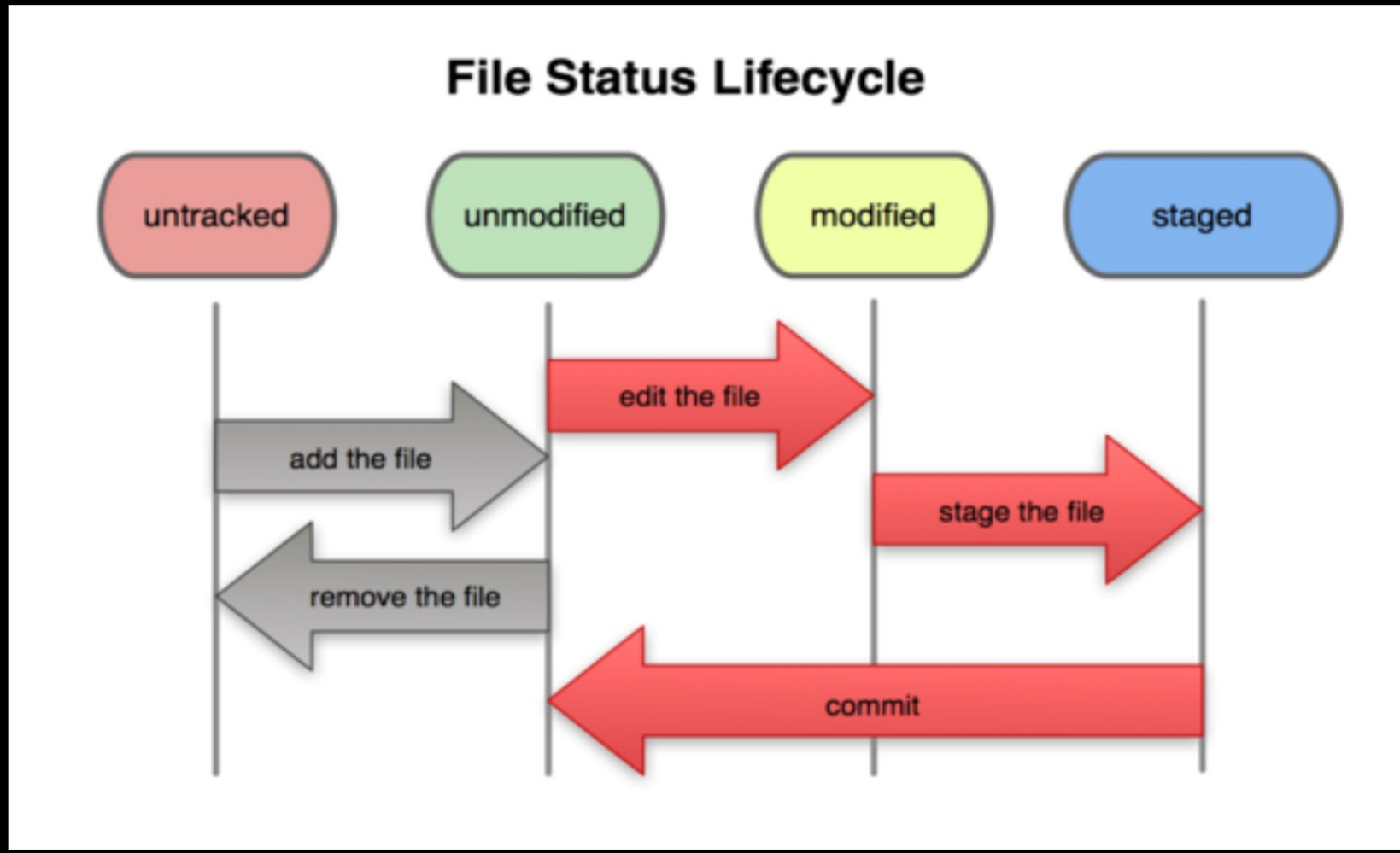
Subversion



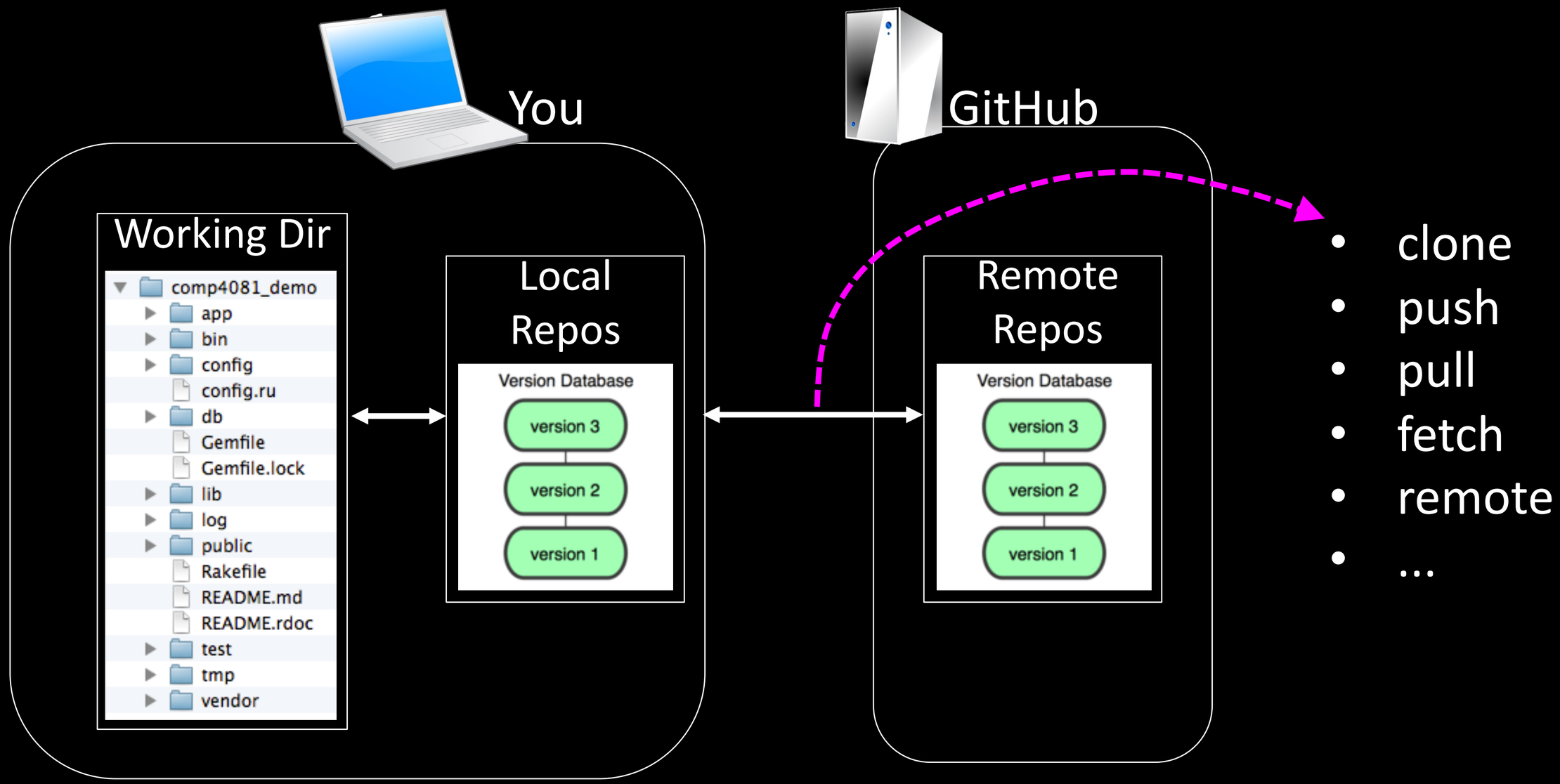
Git




Local Operations
working directory
staging area
git directory (repository)
checkout the project
stage files
commit
Unmodified/modified Files
Staged Files
Committed Files


File Status Lifecycle
untracked
unmodified
modified
staged
edit the file
add the file
stage the file
remove the file
commit

Remote Repo Operations



- clone
- push
- pull
- fetch
- remote
- ...

Git/GitHub

- Install: git :: on your system
- Set your username and email

1. $ git config --global user.name "abc"
2. $ git config --global user.email "abc@gmail.com"

Check your setup

1. $ git config –list

Basic Commands to check status/log

- $ git status
- $ git log
- $ git reflog

Process-1

Set Gihub account

Create your Repository: Test1

Copy Test1 in your local folder(directories) your

$ git clone …..https_link……
$ ls -a

You will find .git file

Add new file or update existing file any file

Go for Staging Area
$ git add .

Go for commit (local git repository commit)
$ git commit -m “first commit”

Push to Github

$ git push origin main

$ git push -u origin main

Process-2

Create local repository first and create one file

Initialize git
$ git init

$ ls -a
You will find .git file

Add new file or update existing file any file

Go for Staging Area
$ git add .

Go for commit (local git repository commit)
$ git commit -m “first commit”

Link to your github repository
$ git remote add origin https://github.com/OWNER/REPOSITORY.git

Check your setup
$ git remote -v

Push to Github

$ git push origin main

$ git push -u origin main

Branching and Merging

Check your branch

- $ git branch

To rename branch

- $ git branch -m main

Create New Branch

- $ git branch newbranchname

Delete any branch

- $ git branch -d branchname

Switch to any branch

- $ git checkout branchname

Branching and Merging

Rest all processes are same for git add/commit/push

Difference between two branches (Let suppose we are on branch: jklu)
We can merge using pull request button over GitHub

- $ git diff main
- $ git merge main

We want the changes done in main after pull request in GitHub to be appeared in our local repo then we need to fetch/download changes in our local repo using this command

- $ git fetch origin main
- Purpose: Update your local copy of remote branches without touching your files

- $ git pull origin <branch name>

- $ git pull origin main

- Purpose: Bring remote changes into your current branch immediately

internally = git fetch + git merge (default)
modifies working directory
may create merge commits or conflicts



Thank You

